

15. Manufacturing Production Monitoring System

Objective:

Develop a complete enterprise-grade application using layered architecture with PostgreSQL, ASP.NET Core Web API, and Angular frontend.

Core Modules:

- Production Planning
- Machine Monitoring
- Defect Tracking

Key Features:

- IoT simulation
- Production analytics

Suggested Architecture:

Angular Frontend → ASP.NET Core Web API → Service Layer → Repository Layer → PostgreSQL Database

Azure Integration:

- Azure IoT simulation
- Azure Service Bus

Student Deliverables:

- Requirement Document
- ER Diagram
- API Documentation
- Angular Screens
- Authentication & Authorization
- CRUD APIs
- Search & Pagination
- Deployment Steps

Phase 2 Enhancements:

- Kubernetes